Electronic Circuit Design

Lesson 3: Circuits with Resistive Feedback

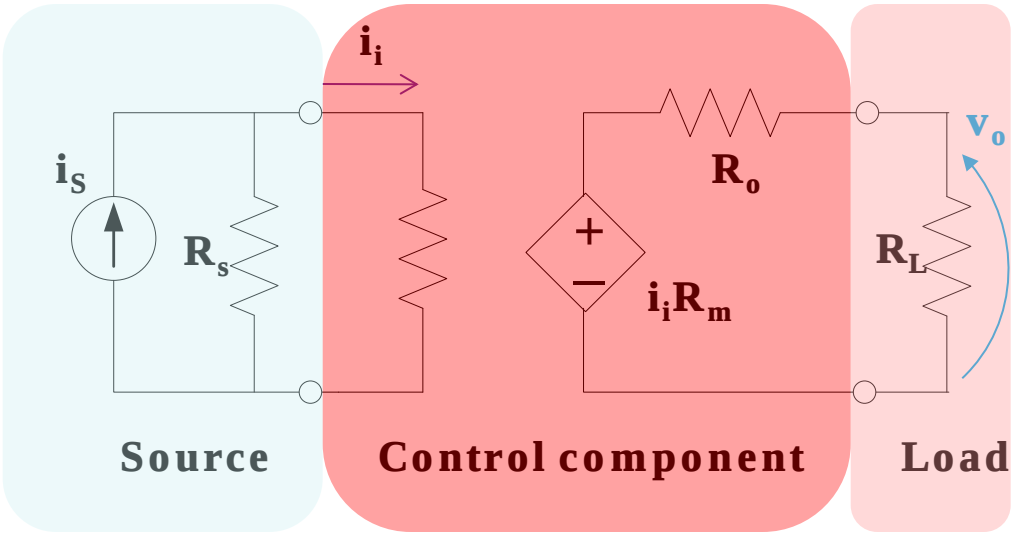
Key Terms

- voltage-to-voltage converter (voltage amplifier)
- current-to-voltage converter
- voltage-to-current converter
- current-to-current converter (current amplifier)
- difference amplifier
- instrumentation amplifier

Current to Voltage Converters

Current to Voltage Converters

A current to voltage converter (I-V converter), also called a transresistance amplifier, which amplifies an input current signal and provide an output voltage signal.



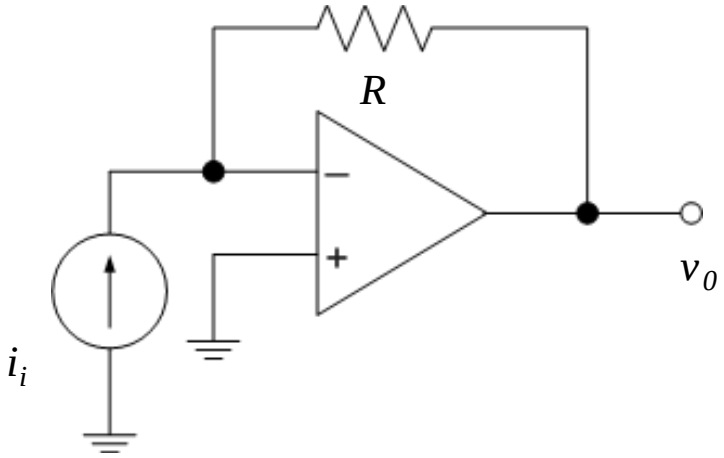
Amplifier ($R_m i_i$)

$$R_m = A_i R_o = A_v R_i$$

Current to voltage converters

Use op amp as the control component with current input and voltage output. The typical circuit shows as below.

Assuming an ideal op amp, apply KCL at v_N gate.



$$i_i + \frac{v_o - 0}{R} = 0 \Rightarrow v_o = -i_i R$$

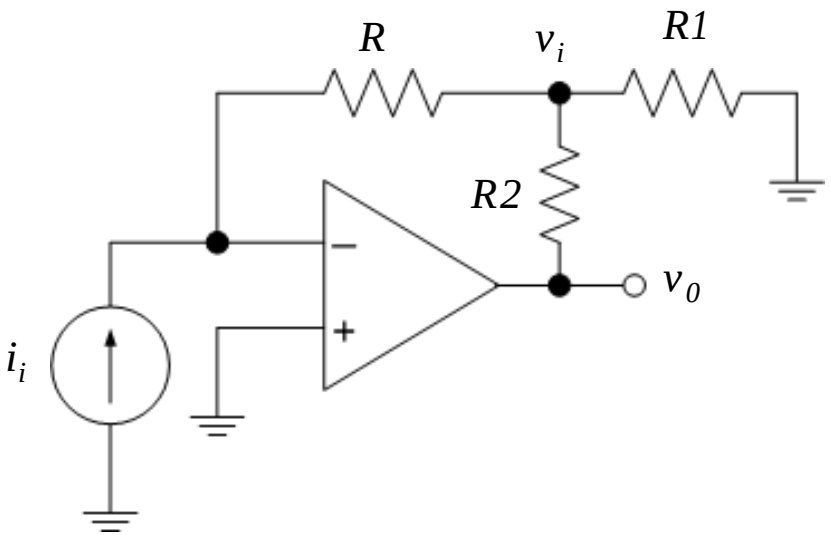
R ≡ transresistance or sensitivity ≡V/A (ohm)

For a small input current, increasing gain (R) to obtain optimal output voltage.

High-sensitivity I-V Converter

If we increase R in order to obtain high sensitivity, R might become impractically large.

Solution: use a T-network to synthesize a large resistance using these relatively small resistors.



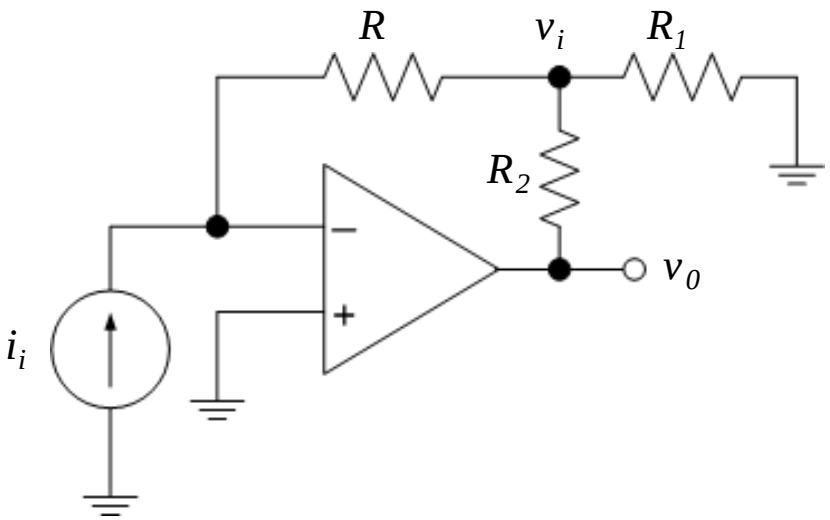
Now, use KCL at v_i

$$\frac{(v_i - 0)}{R} + \frac{(v_i - v_o)}{R_2} + \frac{(v_i - 0)}{R_1} = 0 \quad (1)$$

Also, apply KCL at v_N

$$-i_i - \frac{(v_i - 0)}{R} = 0 \Rightarrow v_i = -i_i R \quad (2)$$

High-sensitivity I-V Converter



Substituting (2) into (1), yields

$$v_o = -i_i R(1 + \frac{R_2}{R} + \frac{R_2}{R_1})$$

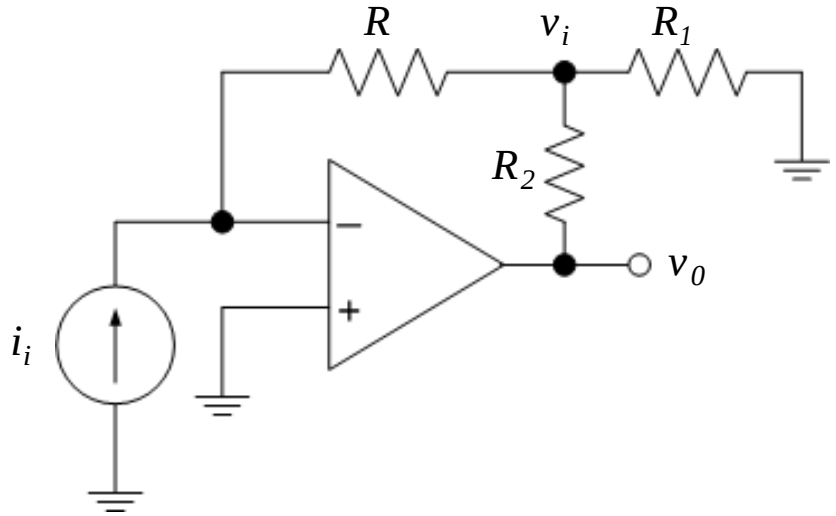
Additional gain (κ multiplier)

$$k = (1 + \frac{R_2}{R} + \frac{R_2}{R_1})$$

A way to obtaining higher gain (transresistance) with lower values of resistors.

High-sensitivity I-V converter

Example: In the circuit, specify suitable component values to achieve a sensitivity of 0.1 V/nA.



High-sensitivity I-V converter

Example: In the circuit, specify suitable component values to achieve a sensitivity of 0.1 V/nA.

Solution

Use $v_o = -KRi_r \Rightarrow \text{sensitivity} = \frac{v_0}{ii} = KR$

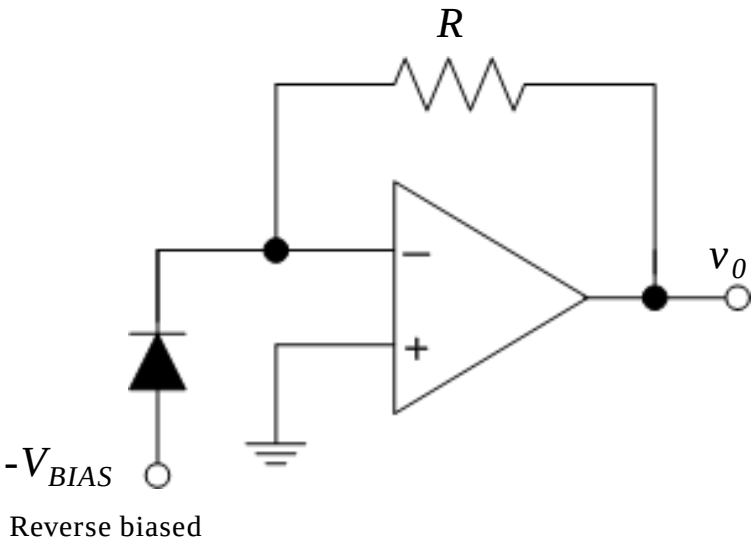
Where $K = 1 + \frac{R_2}{R_1} + \frac{R_2}{R}$; $KR = \frac{0.1}{10^{-9}} = 100Mohm$

Choose $R = 1M\Omega$, then $1 + \frac{R_2}{R_1} + \frac{R_2}{R} = 100$.

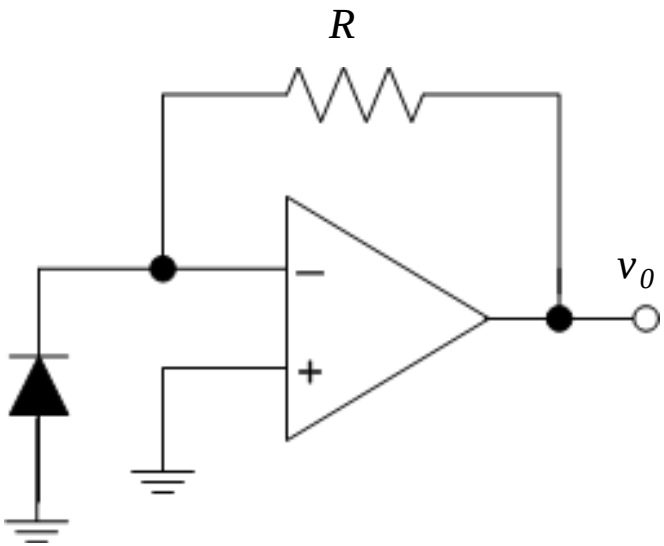
Pick a typical value for $R_1 = 1K\Omega$, then $R_2 \approx 99K\Omega$.

Typical applications of I-V converter

Photodetector amplifiers



Photoconductive



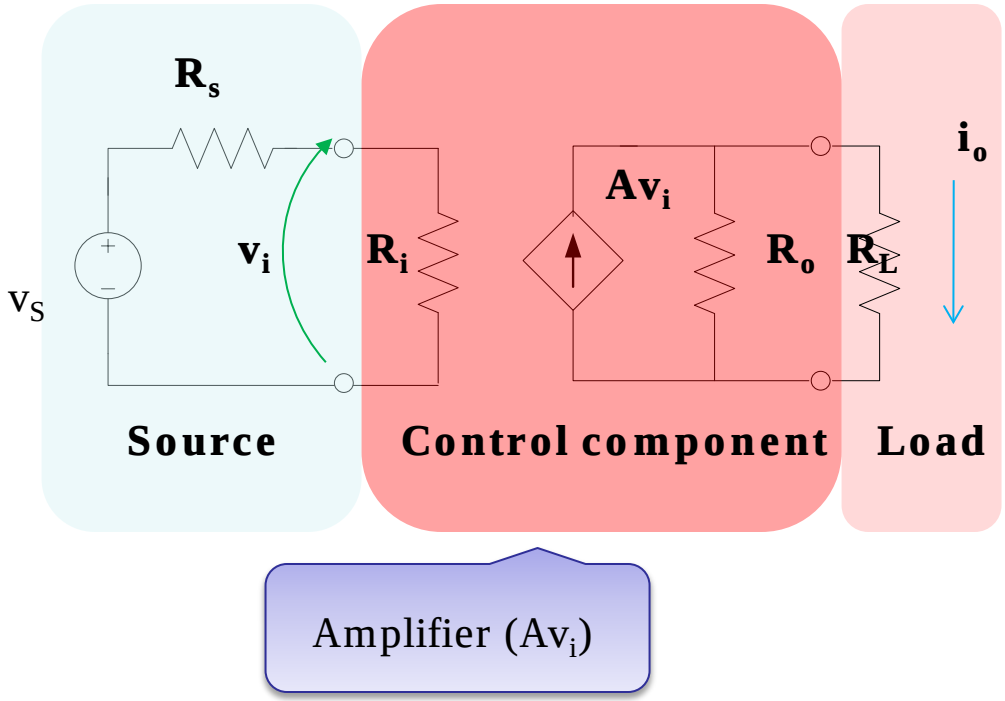
Photovoltaic

Photoconductive mode offers higher speed, and photovoltaic mode offers lower noise such as the dark current.

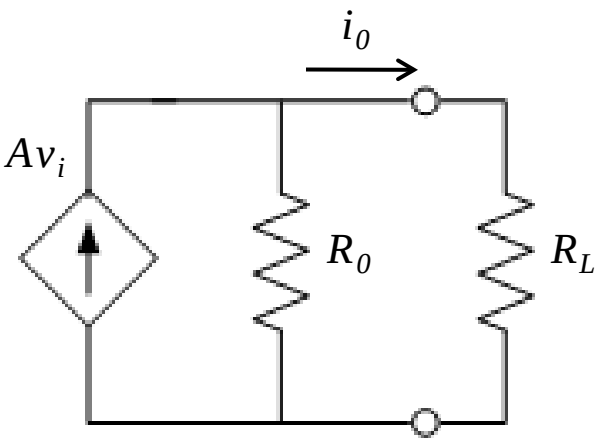
Voltage to Current Converters

Voltage to Current Converter

A voltage to current converter (V-I converter), also called a transconductance amplifier, which amplifies an input voltage signal and provide an output current signal.



Voltage to Current Converter



Ideally,

$$i_o = Av_i \Rightarrow A = \frac{i_o}{v_i} \left(\frac{A}{V}\right)$$

Practically,

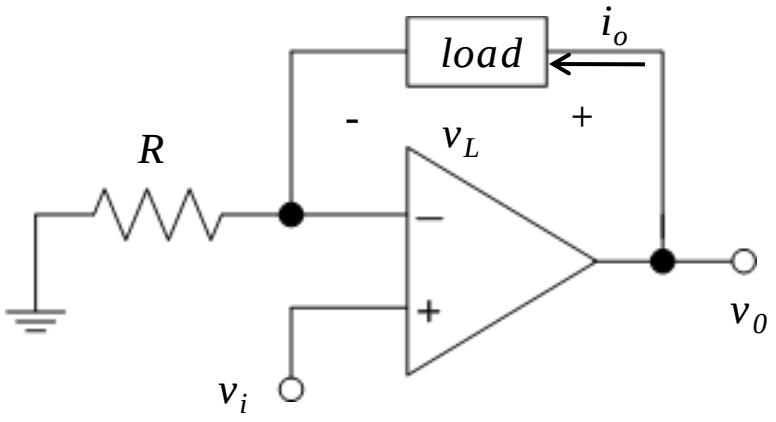
$$i_o = Av_i - {}^{v_L}/_{R_o}$$

v_L: voltage developed by output load
R_O: converter's output resistance seen by load

For true *V-I* conversion, i_O must be independent of v_L; that is, we must have $R_O = \infty$.

Voltage to Current Converter

Floating-load V-I converters A



By virtual short circuit

$$v_N = v_P = v_i$$

Now, apply KCL at

$$\frac{(v_i - 0)}{R} - i_o = 0 \Rightarrow i_o = \frac{v_i}{R}$$

The current is independent on load

Output voltage of op amp:

$$v_o = v_L + v_i \Rightarrow v_L = v_o - v_i$$

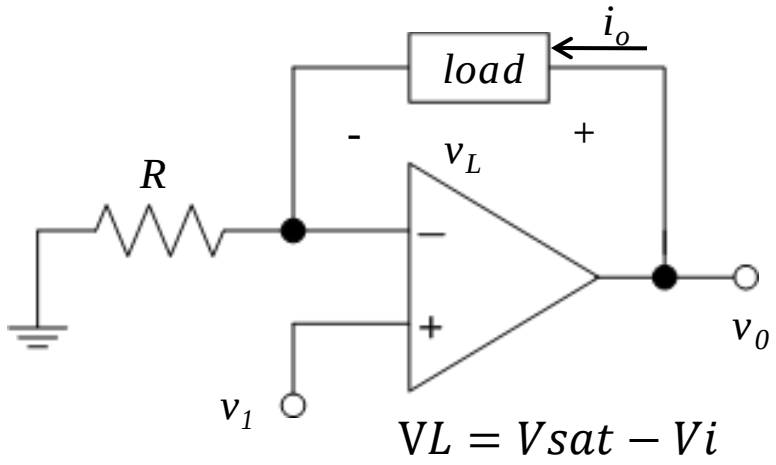
And output swing of the op amp: less than full swing of op amp

$$V_{OL} < v_O < V_{OH}$$

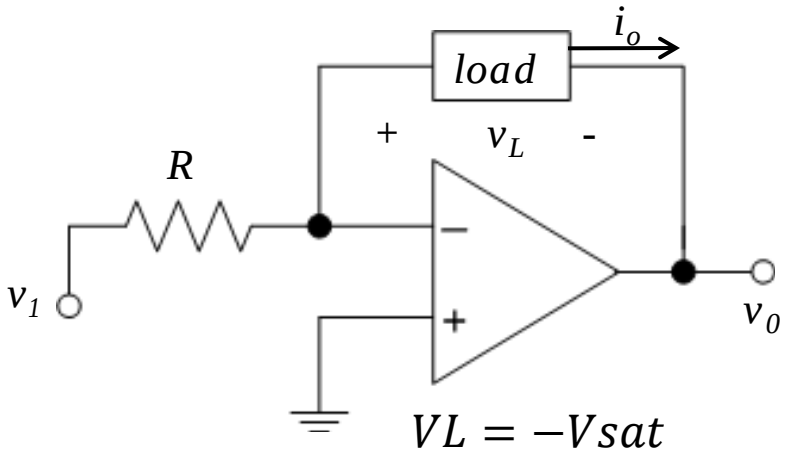
Therefore, the swing limits of v_L, depends on Vi:

$$v_{oL} - v_i < v_L < v_{oH} - v_i$$

Voltage to Current Converter – which one is better?



Floating-load V-I converters A



Floating-load V-I converters B

Tradeoff:

- v_i sees very large input impedance in A, but lower output voltage compliance (swing)
- Full voltage compliance in B, but v_i must provide the load current in addition to op amp.

Voltage to Current Converter

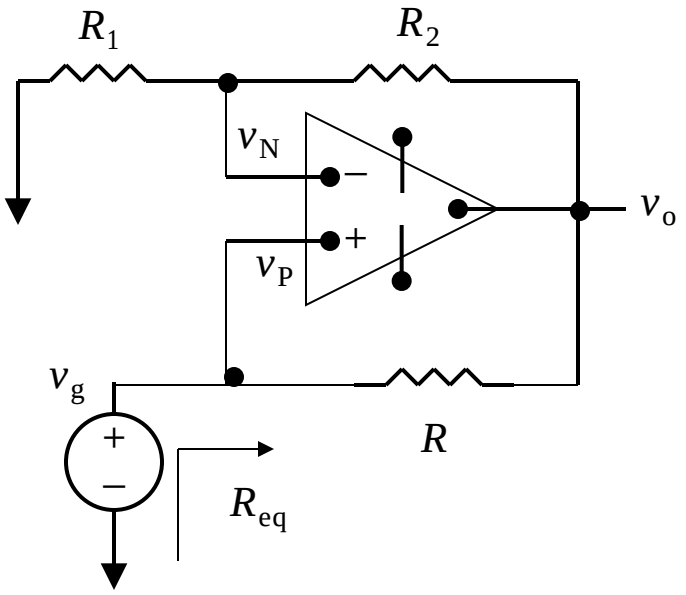
Ground-load V-I converters

Recall: Negative-resistance converter, which provides *impedance transformation*.

From non-inverting characteristics,

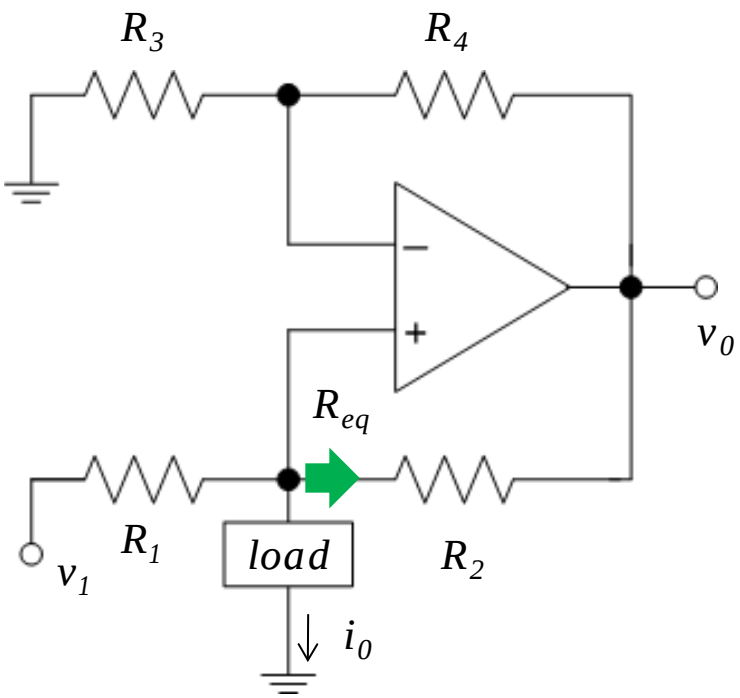
$$v_o = v_g(1 + \frac{R_2}{R_1})$$

$$R_{eq} = \frac{v_g}{i} = -\frac{R_1}{R_2}R$$



Voltage to Current Converter

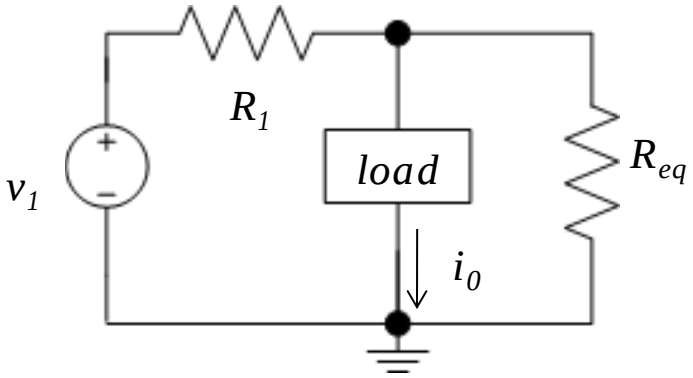
Ground-load V-I converter (Howland current pump)



From NIC configuration, (previous slide)

$$R_{eq} = -\frac{R_3}{R_4} R_2$$

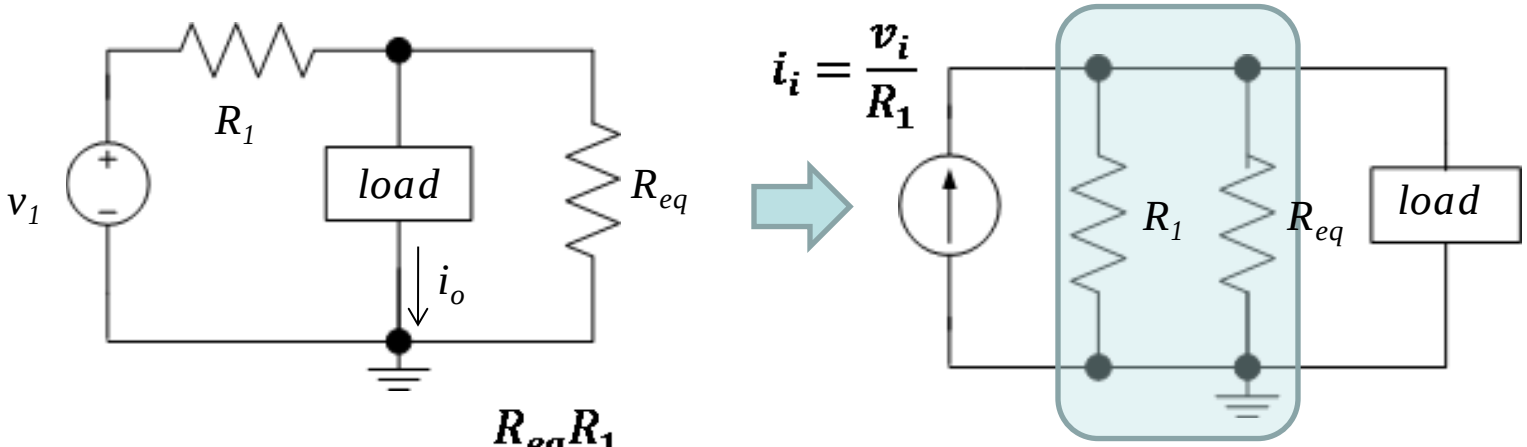
Draw the equivalent circuit,



Voltage to Current Converter

Ground-load V-I converter (Howland current pump)

Now, apply source transformation on voltage source,



$$R_O = R_{eq}//R_1 = \frac{R_{eq}R_1}{R_{eq}+R_1}$$

From $R_{eq} = -\frac{R_3}{R_4}R_2$

This yields $R_O = \frac{R_1(-\frac{R_3}{R_4}R_2)}{-\frac{R_3}{R_4}R_2 + R_1} = \frac{R_2}{R_2/R_1 - R_4/R_3}$

Voltage to Current Converter

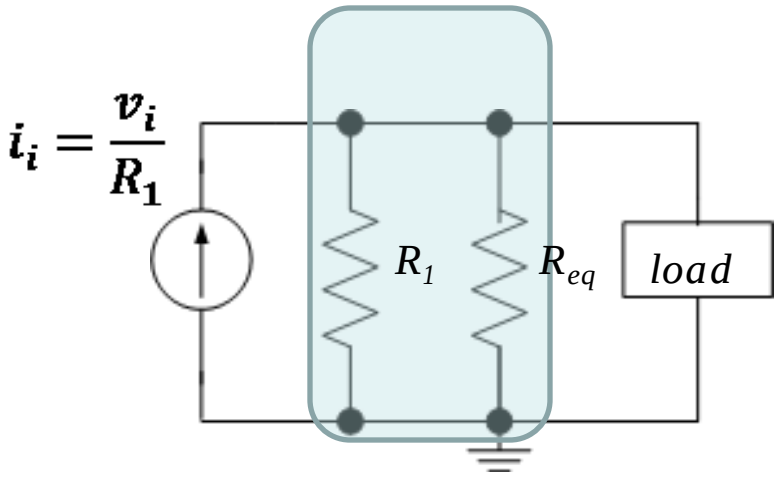
Ground-load V-I converter (Howland current pump)

For a true current source behavior, $R_O \rightarrow \infty$

$$R_o = \frac{R_1\left(-\frac{R_3}{R_4}R_2\right)}{-\frac{R_3}{R_4}R_2 + R_1} = \frac{R_2}{R_2/R_1 - R_4/R_3}$$

Therefore, $R_2/R_1 = R_4/R_3$

Hence, the four resistors form a balanced bridge

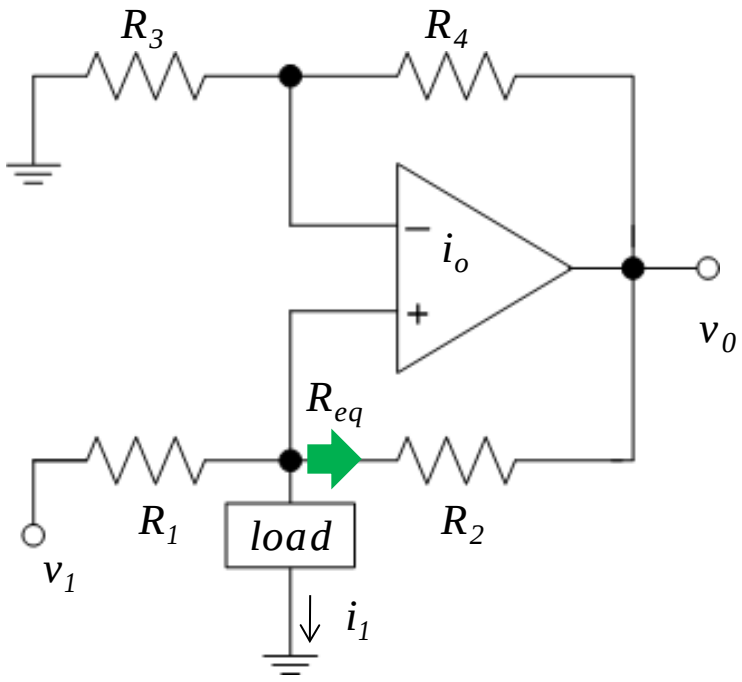


In this case, $i_o = i_i = \frac{v_i}{R_1}$

No loss through source resistance

Voltage to current converter

Ground-load V-I converter (Howland current pump)



Using the virtual short circuit behavior,

$$v_N = v_P = \frac{R_3}{R_3 + R_4} v_O = v_L$$

Hence, the voltage compliance is

$$|v_L| \leq \frac{R_3}{R_3 + R_4} v_{sat} \; or \; \frac{R_1}{R_1 + R_2} v_{sat}$$

The output swing of the op amp is symmetric with respect to zero voltage, independent of Vi.

Voltage to Current Converter

Example: Using a 741 op amp powered from ± 15 V regulated supplies, design a 1-mA dc source having a 10-V voltage compliance. Vi = 15V (use Howland current pump).

Voltage to Current Converter

Example: Using a 741 op amp powered from ± 15 V regulated supplies, design a 1-mA dc source having a 10-V voltage compliance. Vi = +15V.

Voltage to Current Converter

Example: Using a 741 op amp powered from ± 15 V regulated supplies, design a 1-mA dc source having a 10-V voltage compliance. Vi = +15V ; Vsat = +/- 13V.

Solution:

From $i_o = \frac{v_i}{R_1} \Rightarrow R_1 = \frac{v_i}{i_o} = \frac{15V}{1mA} = 15\ k\Omega$

And voltage compliance: $|v_L| = \frac{R_3}{R_3 + R_4}|V_{OH}|$

Therefore: $10V = 13V\,\frac{R_3}{R_3 + R_4}\,(V_{OH} = V_{sat} = 13V\ for\ 741)$

That is: $R_4 = 0.3R_3$ $\quad R_2 = 0.3R_1$

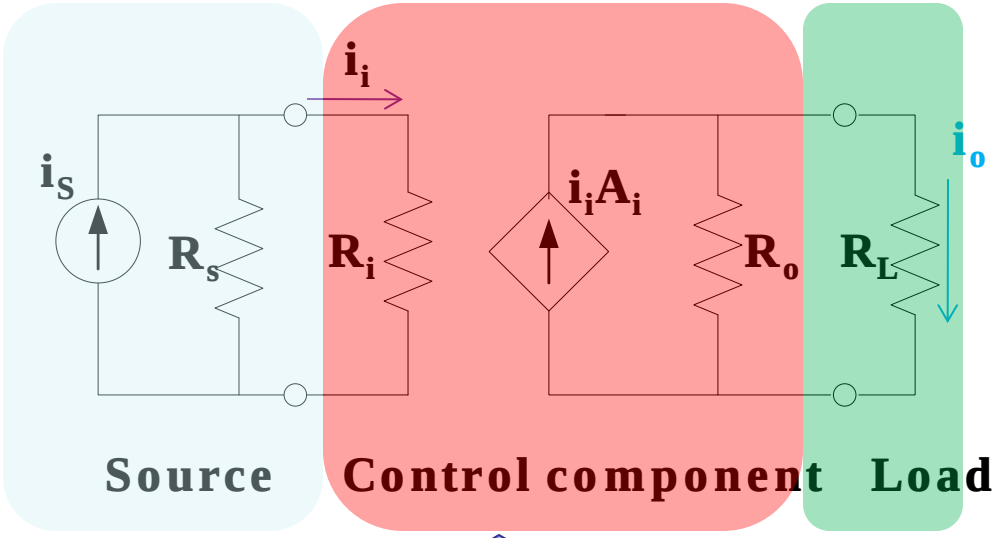
Pick: $R_1 = R_3 = 15K\Omega$, $R_2 = R_4 = 4.5K\Omega$, since $R_2/R_1 = R_4/R_3$

(Use 4.42 KΩ 1%).

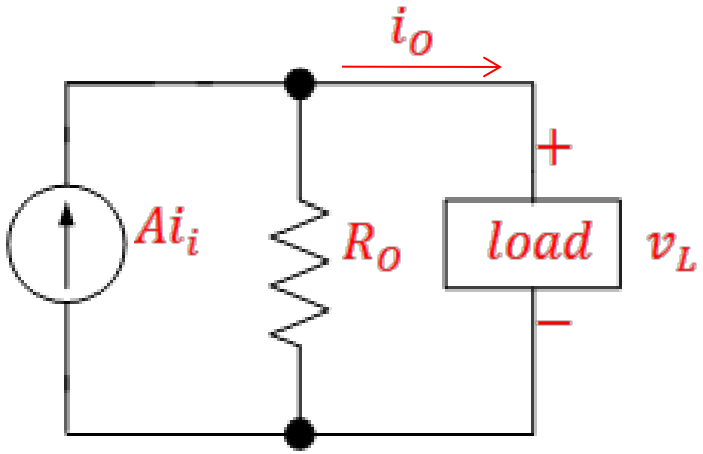
Current Amplifiers

Current amplifier

Current Amplifier: to amplify an input current signal and provide an output current signal



Amplifier ($A_i i_i$)



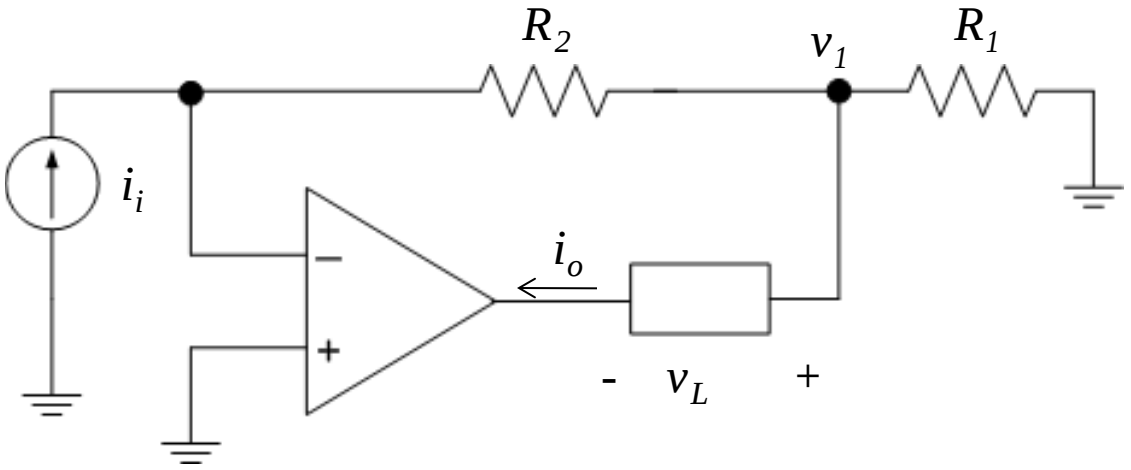
$$i_o = Ai_i - {}^{v_L}/_{R_O}$$

$i_O = Ai_i$ is independent of v_L if $R_O \rightarrow \infty$.

Current Amplifier – Find io/ii

Floating Load

Use ideal op amp as an example,



Apply KCL on v_1:

$$\frac{v_1 - 0}{R_2} + \frac{v_1 - 0}{R_1} + i_o = 0 \quad \text{and} \quad v_1 = -i_i R_2$$

This yields $$v_1 \left[\frac{1}{R_1} + \frac{1}{R_2}\right] = -i_o$$

Therefore, $$i_o = (1 + \frac{R_2}{R_1}) i_i$$

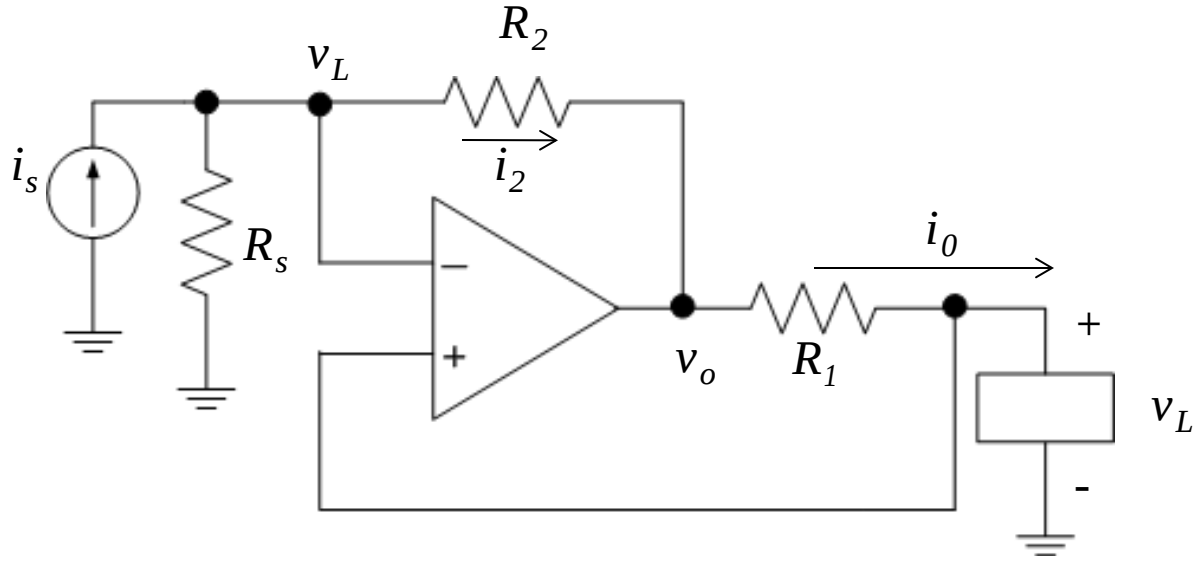
$(1+R_2/R_1)$ is current gain

Current Amplifier

Ground Load

Use ideal op amp as an example:

$v_N = v_P = v_L$



and

$$i_o = \frac{v_o - v_L}{R_1}; \quad i_2 = \frac{v_L - v_o}{R_2}$$

This yields $i_o = -\frac{R_2}{R_1} i_2$

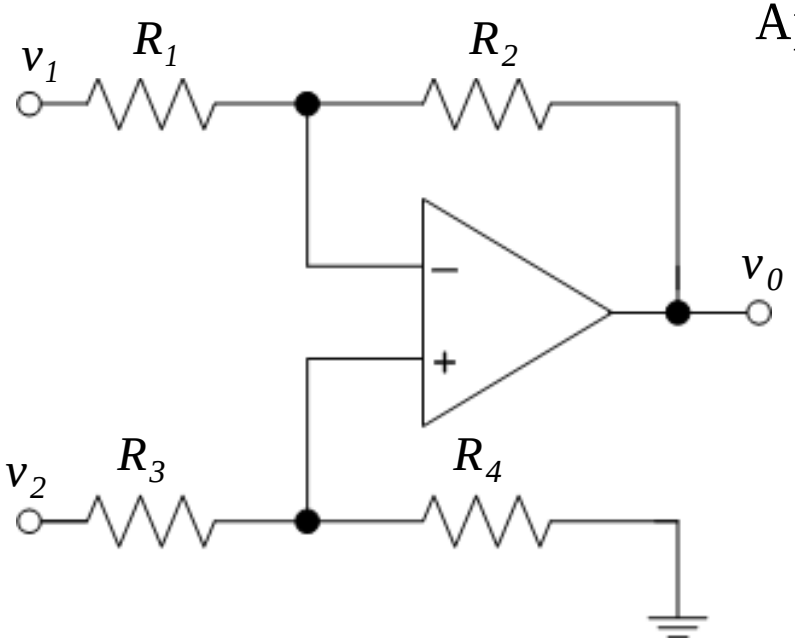
As $R_s \to \infty$, $i_S = i_2$ and $-R_2/R_1$ *is current gain*

$$\frac{i_o}{i_s} = -\frac{R_2}{R_1}$$

Negative current gain means current flows in opposite direction as shown. Hence, with $R_1 = R_2$, the circuit reverses the current flow.

Difference Amplifiers

Difference Amplifiers



Apply superposition to solve the circuit,

$$v_1 \neq 0 \text{ and } v_2 = 0$$

This yields

$$v_0' = -\frac{R_2}{R_1} v_1$$

And $v_1 = 0 \text{ and } v_2 \neq 0$

$$v_0'' = \frac{R_4}{R_3 + R_4} v_2 \times \left(\frac{R_1 + R_2}{R_1}\right)$$

Therefore,

$$v_O = -\frac{R_2}{R_1} v_1 + \frac{R_4}{R_3 + R_4} v_2 \times \left(\frac{R_1 + R_2}{R_1}\right)$$

Difference amplifiers are very important circuits, based on a balanced bridge configuration.

Difference Amplifier

Ex:

If (V1,V2) = (9.9v, 10.1v)

a) Find Vo if R1=10k, R2=98k, R3=9.9k, R4=103k.

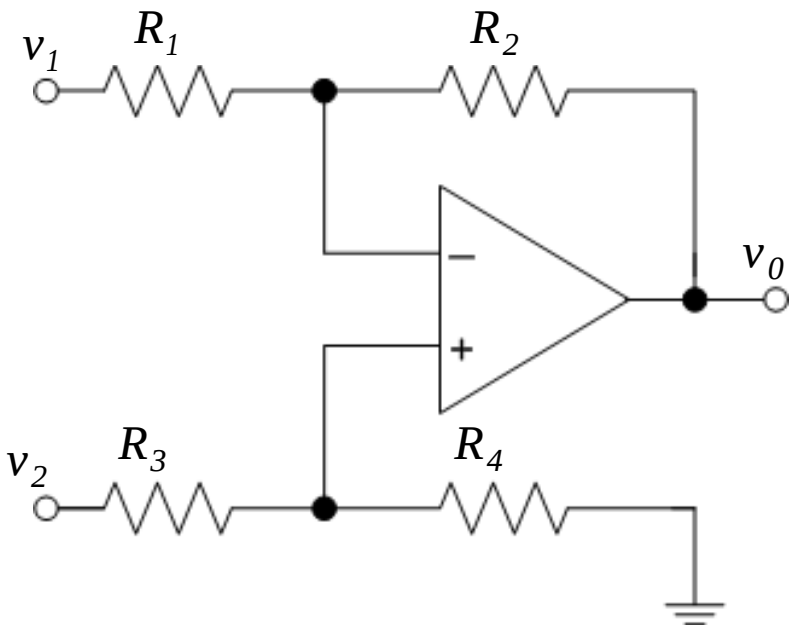
b) Find Vo if R1=R3=10k and R2=R4=100k

Ans:

a) Vo=2.495v

b) Vo=2v

Difference Amplifiers



$$v_O = -\frac{R_2}{R_1}v_1 + \frac{R_4}{R_3+R_4}v_2 \times \left(\frac{R_1+R_2}{R_1}\right)$$

$$= \frac{R_2}{R_1}\left[\frac{R_4(R_1+R_2)}{R_2(R_3+R_4)}v_2 - v_1\right]$$

Designed output voltage has the form

$$v_O = \frac{R_2}{R_1}(v_2 - v_1)$$

Choose, R_4=R_2 and R_3=R_1 to satisfy the balanced-bridge condition.

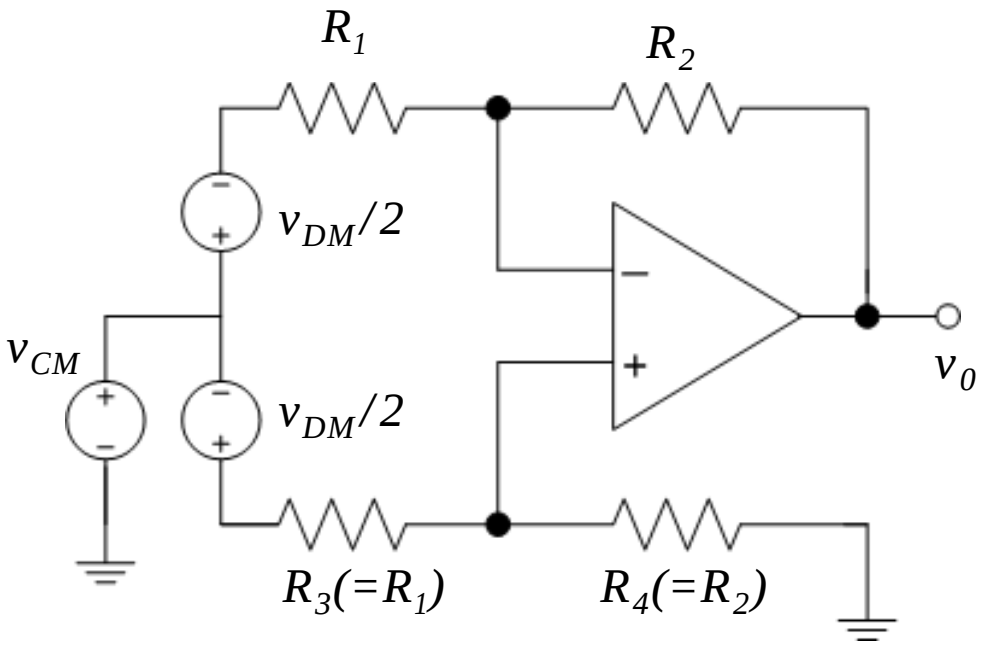
Specify inputs in terms of common-mode and differential-mode components:

$$v_{DM} = v_2 - v_1 \quad \text{and} \quad v_{CM} = \frac{v_1+v_2}{2}$$

Difference Amplifiers

$$v_{DM} = v_2 - v_1 \quad \text{and } v_{CM} = \frac{v_1 + v_2}{2}$$

$$\Rightarrow v_1 = v_{CM} - \frac{v_{DM}}{2} \text{ and } v_2 = v_{CM} + \frac{v_{DM}}{2}$$



For pure *CM* input voltage only,

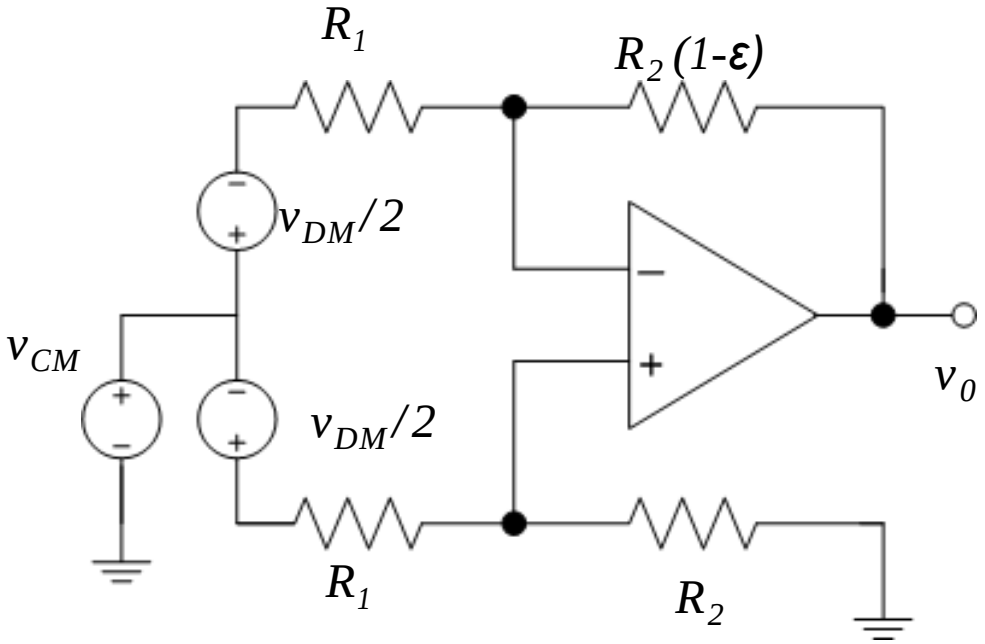
$$v_o = \frac{R_2}{R_1}(v_2 - v_1)$$

$$= \frac{R_2}{R_1}(v_{CM} - v_{CM}) = 0$$

An ideal differential amplifier fully rejects common-mode input signals.

Difference Amplifiers

Effects of resistance mismatches



Previously, we have

$$v_o = -\frac{R_2}{R_1}v_1 + \frac{R_4}{R_3+R_4}v_2 \times \left(\frac{R_1+R_2}{R_1}\right)$$

Apply superposition principle,

$$v_o = -\frac{R_2(1-\varepsilon)}{R_1}\left(v_{CM} - \frac{v_{DM}}{2}\right) +$$

$$\frac{R_2}{R_1+R_2}\left(v_{CM} + \frac{v_{DM}}{2}\right) \times \frac{R_1+R_2(1-\varepsilon)}{R_1}$$

Multiplying and collecting terms,

$$v_o = \boxed{\frac{R_2}{R_1}\left[\frac{1-\varepsilon}{2} + \frac{R_1+R_2(1-\varepsilon)}{2(R_1+R_2)}\right]}v_{DM} + \boxed{\left[\frac{R_2}{R_1+R_2}\frac{R_1+R_2(1-\varepsilon)}{R_1} - \frac{R_2(1-\varepsilon)}{R_1}\right]}v_{CM}$$

A_{DM} A_{CM}

Difference Amplifiers

$$v_O = \frac{R_2}{R_1}\left[\frac{1-\varepsilon}{2} + \frac{R_1 + R_2(1-\varepsilon)}{2(R_1+R_2)}\right] v_{DM} + \left[\frac{R_2}{R_1+R_2}\,\frac{R_1 + R_2(1-\varepsilon)}{R_1} - \frac{R_2(1-\varepsilon)}{R_1}\right] v_{CM}$$

A_{DM} A_{CM}

Simplifying, $A_{DM} = \frac{R_2}{R_1}\left(1 - \frac{R_1 + 2R_2}{R_1 + R_2} \times \frac{\varepsilon}{2}\right)$ and $A_{CM} = \frac{R_2}{R_1 + R_2}\varepsilon$

Differential mode gain desired to be large

Common mode gain desired to be small

CMRR: common mode rejection ratio: $20 \log_{10} \left|\frac{A_{DM}}{A_{CM}}\right|$

Difference Amplifiers

Example: LM741 op amp

CMRR (Min) = 70 dB
CMRR (typical) = 90 dB

Note: CMRR (Max) = ∞ if everything happens to match perfectly (V_o = 0 pure V_{CM}).

Recall:

$$A_{DM} = \frac{R_2}{R_1}\left(1 - \frac{R_1 + 2R_2}{R_1 + R_2} \times \frac{\varepsilon}{2}\right)$$

$$A_{CM} = \frac{R_2}{R_1 + R_2}\varepsilon$$

For small ε,

$$A_{DM} \cong \frac{R_2}{R_1} \text{ and } A_{CM} \cong \frac{R_2}{R_1 + R_2}\varepsilon$$

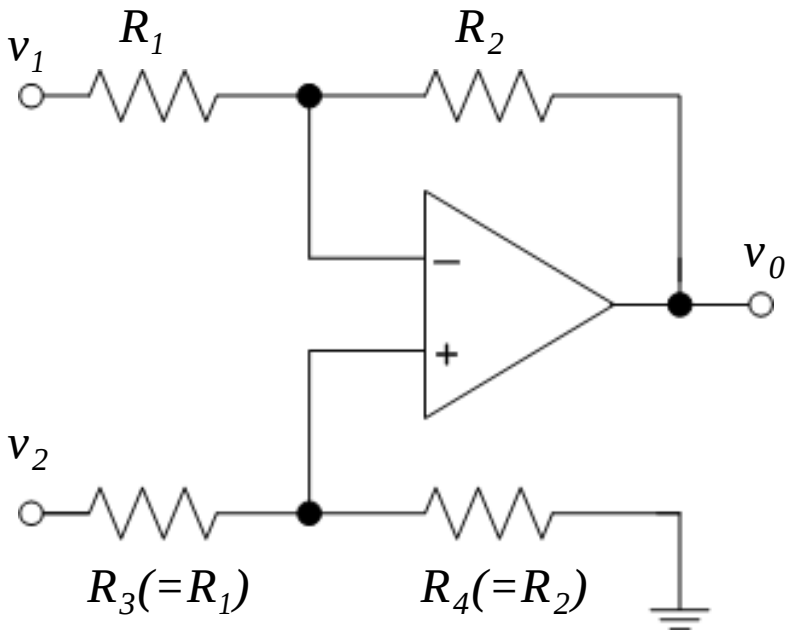
$$\therefore CMRR_{dB} = 20\log_{10}\left|\frac{R_2/R_1}{R_2\varepsilon/R_1 + R_2}\right|$$

$$= 20\log_{10}\left|\frac{1 + R_2/R_1}{\varepsilon}\right|$$

As R_2/R_1 increases, CMRR also increases.

Difference Amplifiers

Variable gain:



Choose, $R_4=R_2$ and $R_3=R_1$ to satisfy the balanced-bridge condition.

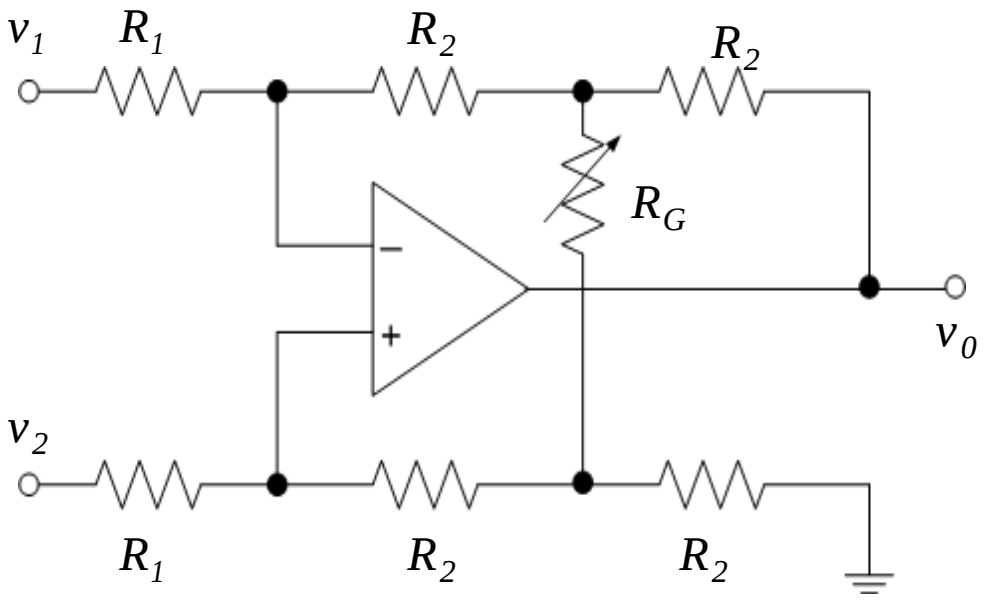
Recall: $v_o = \frac{R_2}{R_1}(v_2 - v_1)$

This suggests that only one resistor needs to be changed to vary the gain. To reach balance, however, we have to keep the two resistors variable.

This is an awkward DM solution

Difference Amplifiers

Variable gain:



The better solution as shown, output voltage:

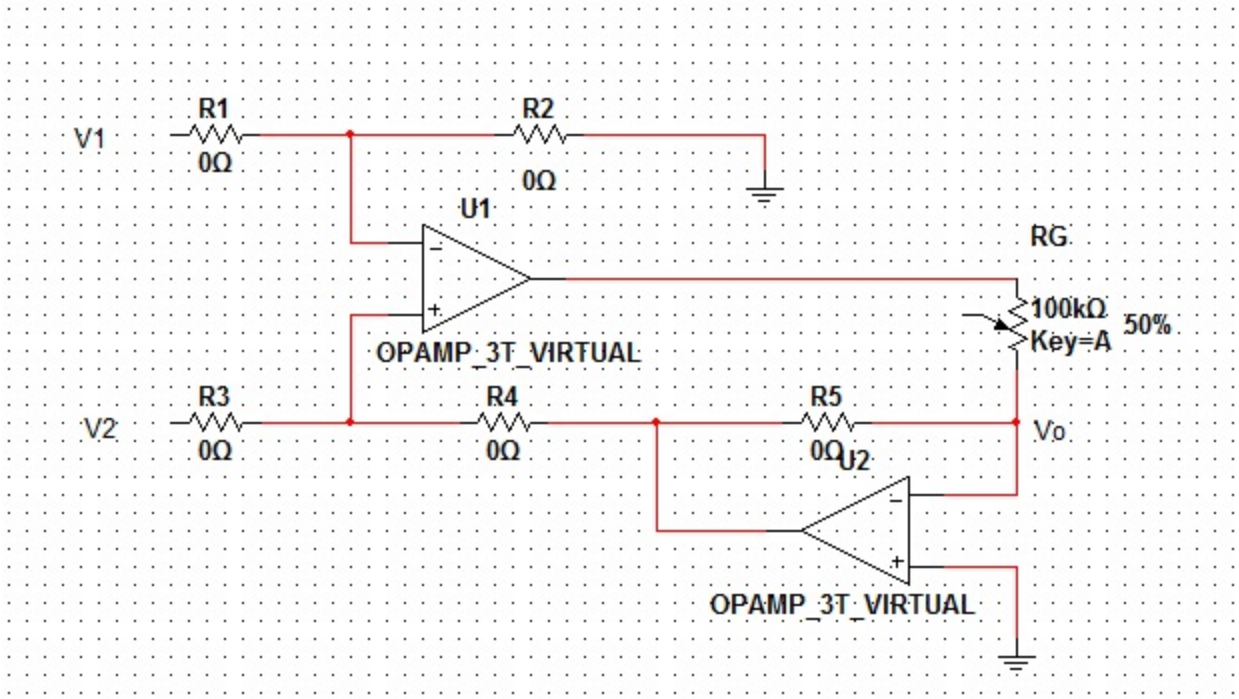
$$v_o = \frac{2R_2}{R_1}\left(1 + \frac{R_2}{R_G}\right)(v_2 - v_1)$$

Gain varied by single resistor R_G

Problems:

- This is a non-linear dependence of gain on RG.
- Adjustable resistor is big, costly, and requires manpower to adjust each device.

Difference Amplifier with Linear Gain Control



$$R_3 = R_1 \qquad R_4 = R_2 \qquad v_o = \frac{R_2 R_G}{R_1 R_5}(v_2 - v_1)$$

Ex

Specify suitable component values such that gain can be varied from 1 V/V to 100 V/V.

(b) Let $R_1 = R_2 = 10\,k\Omega$. Then, $A = R_G/R_3$. Let $R_3 = 1k\Omega$ and let R_G be a 100-$k\Omega$ pot in series with a 1-$k\Omega$ resistor. Then,

$A_{(min)} = 1$ V/V, $A_{(max)} = (1+100)/1 \cong 100$ V/V.

Instrumentation Amplifiers

Instrumentation Amplifiers (IA)

Extremely high R_{id} and R_{ic} impedance

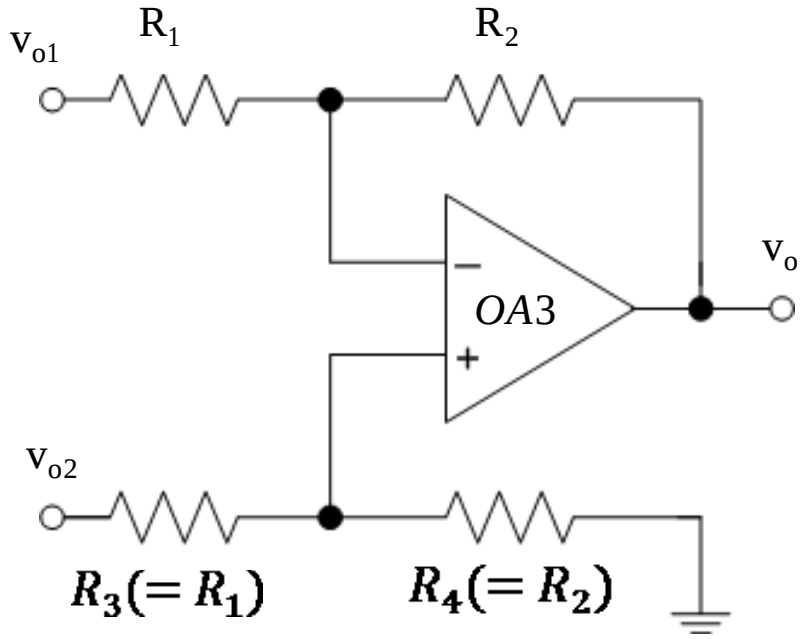
Very low output impedance

Accurate and stable gain

Extremely high CMRR

Instrumentation Amplifiers (IA)

Let's start with a difference amplifier:



Recall,

$$A_{DM} \cong \frac{R_2}{R_1} \text{ and } A_{CM} \cong \frac{R_2}{R_1 + R_2}\varepsilon$$

Differential mode:

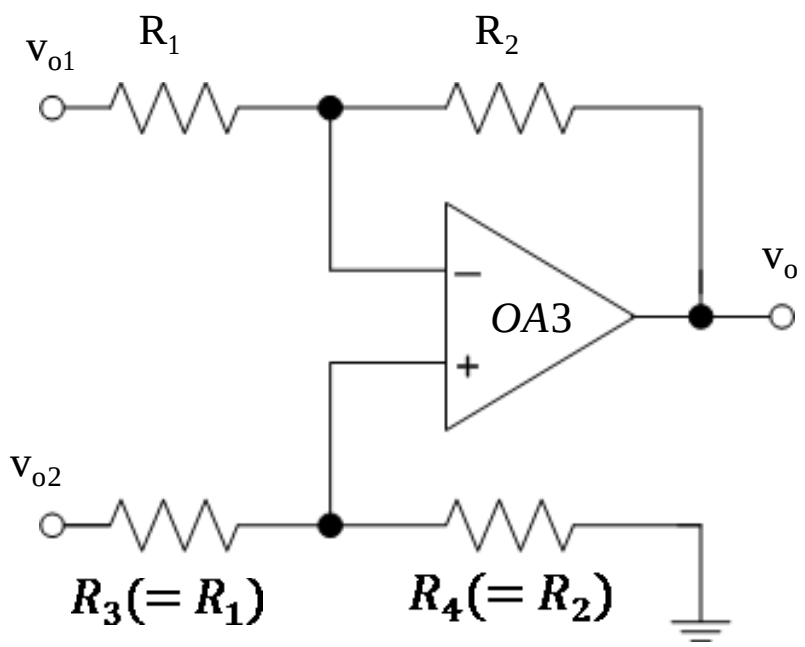
$$v_o = \frac{R_2}{R_1}(v_{o2} - v_{o1})$$

Common mode:

$$v_o = \frac{R_2}{R_1 + R_2}\varepsilon(v_{o2} + v_{o1})$$

$$CMRR_{dB} = 20\log_{10}\left|\frac{1 + {}^{R_2}/_{R_1}}{\varepsilon}\right|$$

Instrumentation Amplifiers (IA)



Challenges: ideal difference amplifier

$$A_{DM} = \frac{R_2}{R_1} \text{ and } A_{CM} = 0$$

Also, $R_i = 2R_1$ Use KVL loop around inputs and Vp and Vn

Input resistance might not large enough for some applications => buffer opamp

Instrumentation Amplifiers (IA)

Triple-Op-Amp IAs: Add voltage followers to input (differential preamp).

$$i_{R_G} = \frac{(v_2 - v_1)}{R_G}$$

$$v_{o2} - v_{o1} = i_{R_G}(R_3 + R_G + R_3)$$

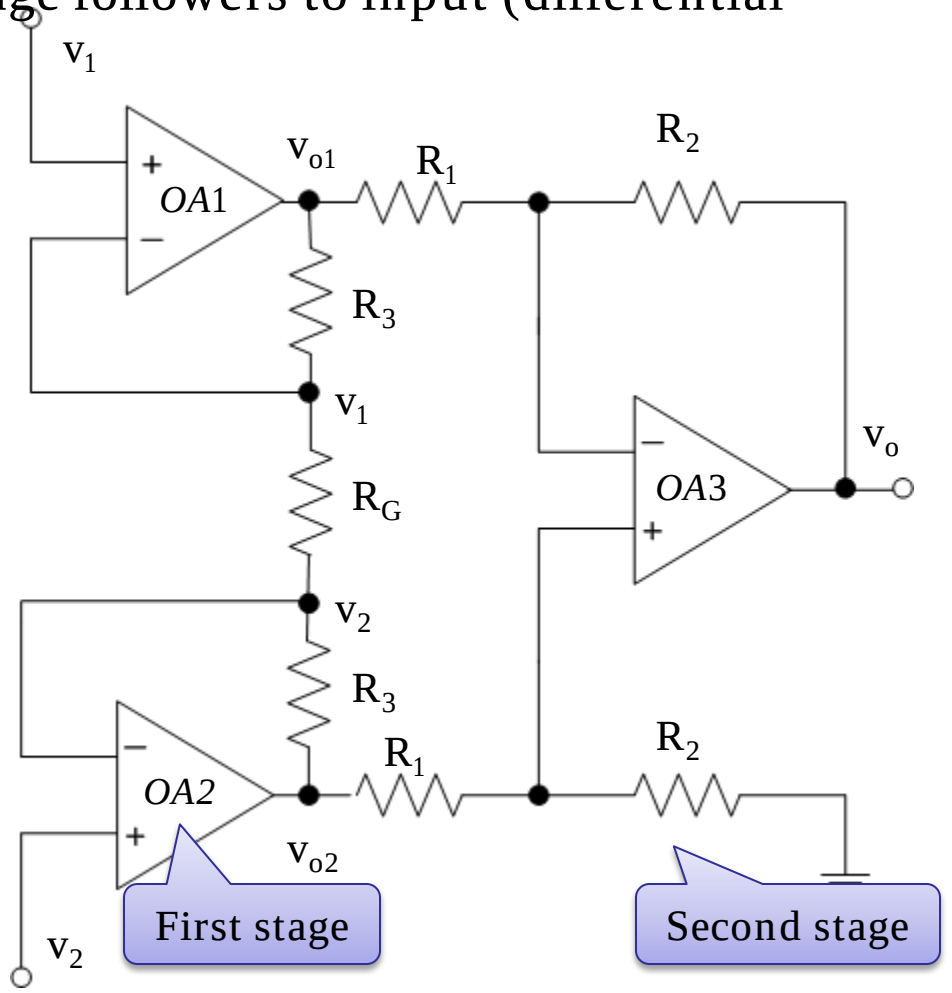
Differential mode (from eqns):

$$v_{o2} - v_{o1} = \left(1 + \frac{2R_3}{R_G}\right)(v_2 - v_1)$$

$$v_o = \frac{R_2}{R_1}(v_{o2} - v_{o1})$$

$$= \frac{R_2}{R_1}\left(1 + \frac{2R_3}{R_G}\right)(v_2 - v_1)$$

Vary R_G to vary gain.



Instrumentation Amplifiers (IA)

Triple-Op-Amp IAs: Add voltage followers to input (differential preamp).

Differential mode:

$$v_{O2} - v_{O1} = \left(1 + \frac{2R_3}{R_G}\right)(v_2 - v_1)$$

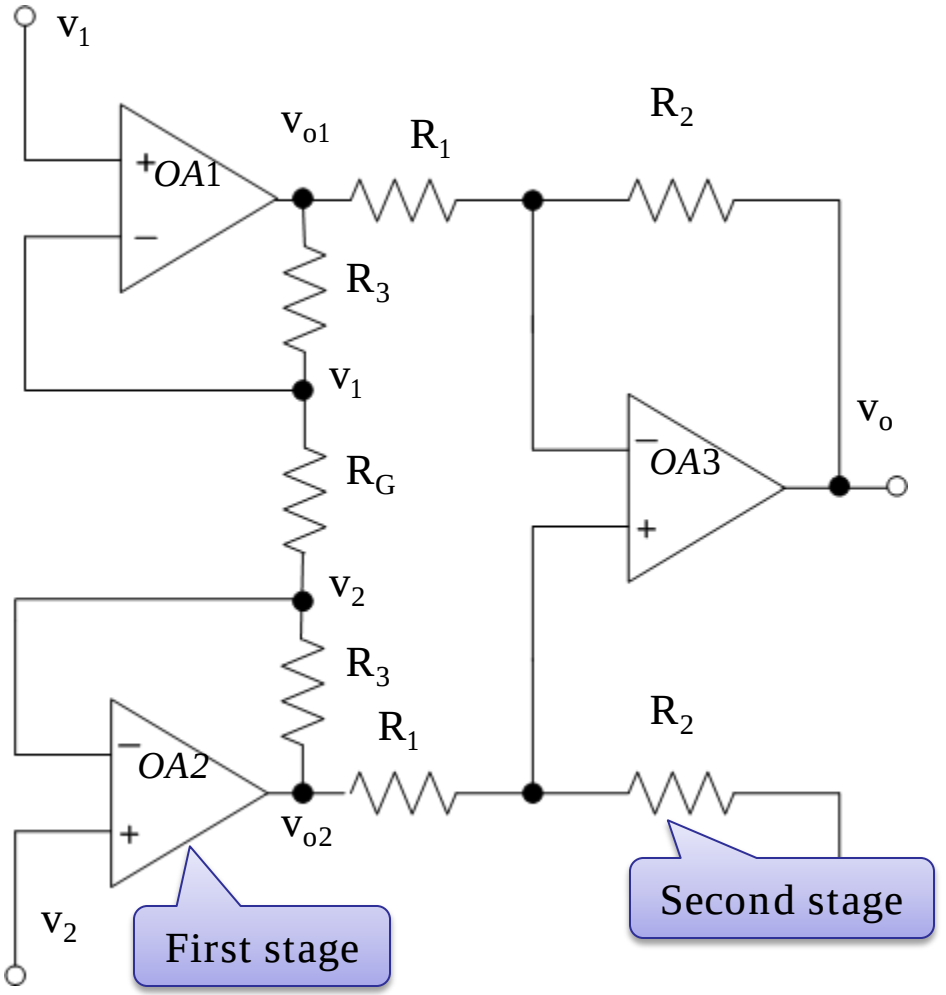
Common mode by def: ½(Vo2+Vo1)
(V1=V2, no Current thro' R3,RG,R3):

$$\frac{1}{2}(v_{o2} + v_{o1}) = 1 \times \frac{1}{2}(v_2 + v_1)$$

$$CMRR = \frac{Adm}{Acm} = \frac{1 + \frac{2R_3}{R_G}}{1/2}$$

$$CMRR = 2(1 + \frac{2R_3}{R_G})$$

CMRR for 1st stage



Instrumentation Amplifiers (IA)

Triple-Op-Amp IAs: Add voltage followers to input (differential preamp),

Differential mode:

$$v_{o2} - v_{o1} = \left(1 + \frac{2R_3}{R_G}\right)(v_2 - v_1)$$

Common mode:

$$\frac{1}{2}(v_{o2} + v_{o1}) = 1 \times \frac{1}{2}(v_2 + v_1)$$

$$CMRR = 2(1 + \frac{2R_3}{R_G})$$

CMRR for 1st stage

$$2nd\ stage\ CMRR = (1 + \frac{R_2}{R_1})\frac{1}{\varepsilon}$$

Combine both stages:

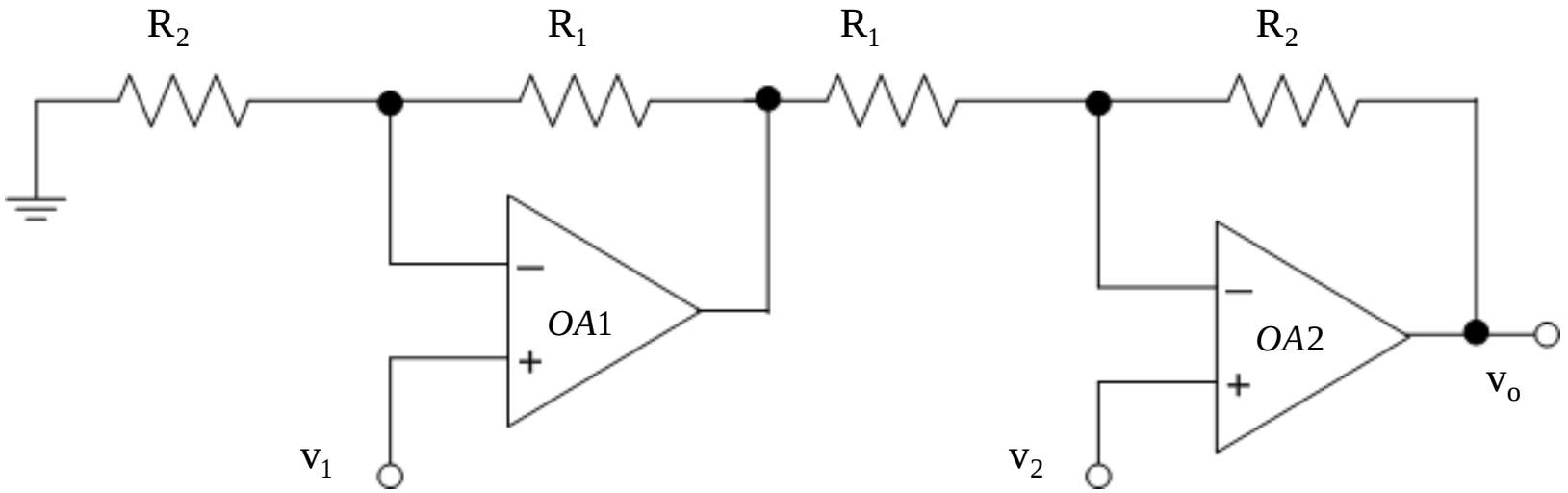
$$CMRR = 2(1 + \frac{2R_3}{R_G})(1 + \frac{R_2}{R_1})\frac{1}{\varepsilon}$$

Can be bigger than differential amplifier

$$\boldsymbol{CMRR = (1 + \frac{R_2}{R_1})\frac{1}{\varepsilon}}$$

Instrumentation Amplifiers (IA)

Dual-Op-Amp IAs:



Now, apply superposition:

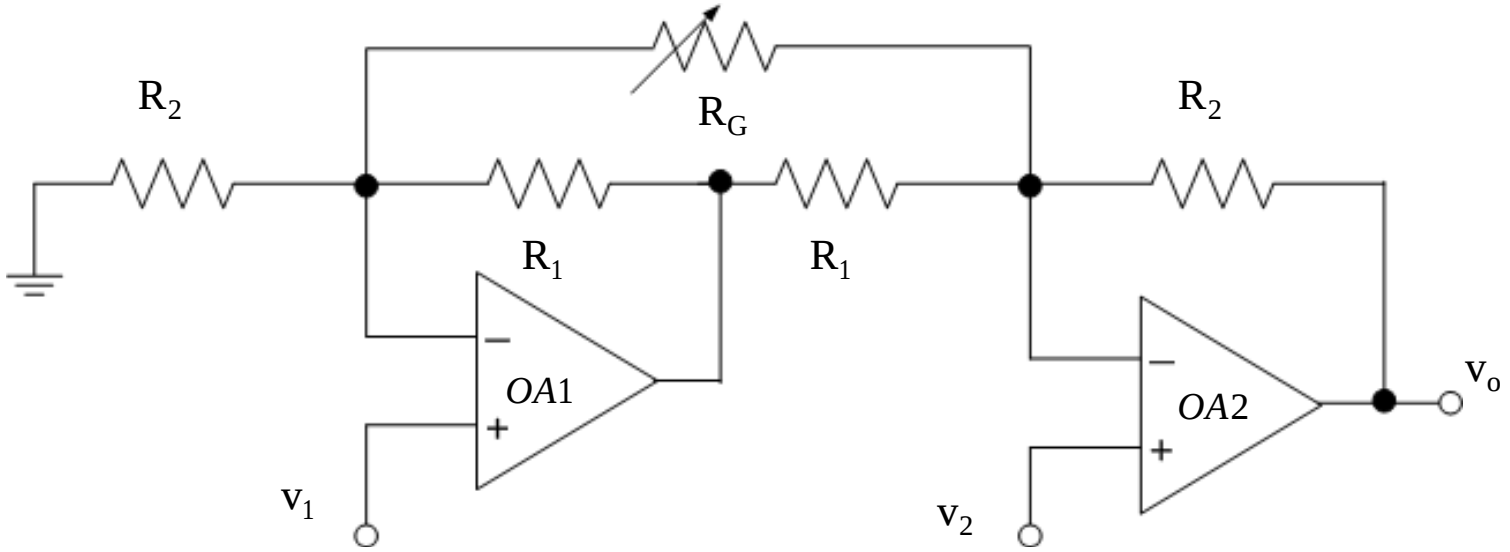
$$v_o = \left(1 + \frac{R_2}{R_1}\right) v_2 \text{ (while } v_1 = 0\text{)}$$

$$-\left(\frac{R_2}{R_1}\right)\left(1 + \frac{R_1}{R_2}\right) v_1 \text{ (while } v_2 = 0\text{)}$$

$$\Rightarrow\ v_o = \left(1 + \frac{R_2}{R_1}\right)(v_2 - v_1)$$

Instrumentation Amplifiers (IA)

Dual-Op-Amp IAs: variable gain

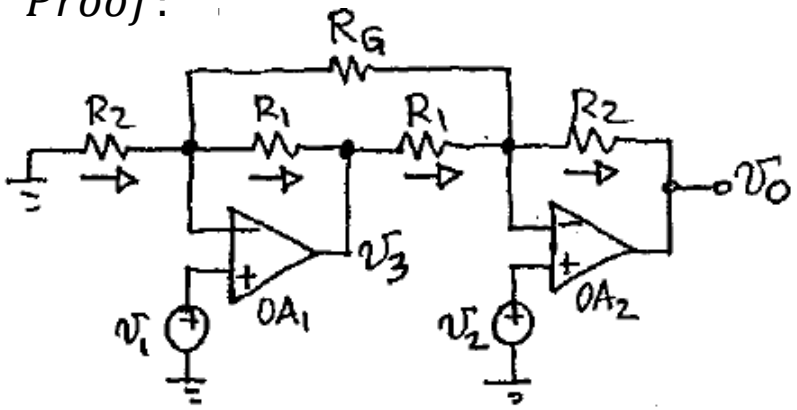


For variable gain with R_G:

$$A = 1 + \frac{R_2}{R_1} + \frac{2R_2}{R_G}$$

Problems include different paths from v_1 and v_2 to v_O. Therefore, inducing higher common mode mismatches at high frequency.

Proof:



$v_{N1} = v_{P1} = v_1$, $v_{N2} = v_{P2} = v_2$. Applying KCL:

$$\frac{0 - v_1}{R_2} = \frac{v_1 - v_2}{R_G} + \frac{v_1 - v_3}{R_1}; \quad \frac{v_2 - v_O}{R_2} = \frac{v_1 - v_2}{R_G} + \frac{v_3 - v_2}{R_1}.$$

Adding the two equations pairwise gives

$$\frac{v_2 - v_1}{R_2} - \frac{v_O}{R_2} = 2\frac{v_1 - v_2}{R_G} + \frac{v_1 - v_2}{R_1}.$$ Solving

for v_O yields $v_O = \left(1 + \frac{R_2}{R_1} + 2\frac{R_2}{R_G}\right)(v_2 - v_1)$.